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MEDIA AND ENVIRONMENTAL FRAMING : A STUDY

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ABSTRACT

Mass media coverage of environmental issues has been recognized as a key element in the policy discourse and public understanding of environment. Media can be given credit for making some of the conceptual, technical terms like 'ozone layer', 'global warming' and 'climate change' as common vocabulary in the public parlance. The aspects of media coverage, science-policy interactions and the public understanding of environment are technically dealt in the circuits of communication model developed. This model illustrates three 'phases' or 'circuits' through which communications originate and disseminate into the public sphere before entering the private sphere. Media Framing is regarded as an important element of the news production phase of the model. In this the news makers i.e. media professionals engaged in the news collection, preparation and writing process construct the news. Thus framing involves selection and salience i.e. to select some aspects of a perceived reality and make them more salient in a communicating text, in such a way as to promote a particular problem definition. Many types of frames like promotion v. prevention, now v. future, gain v. loss, interconnected frame, local frame etc. are used in construction of environmental content by the news makers. The news item analyses the various framing techniques adopted by the media through content analysis of news items published by the regional media in Andhra Pradesh, India.

Key Words : Environmental communication, Environmental framing, Circuits of communication model, News makers, Public understanding of environment

INTRODUCTION

The concept of environmentalism has touched every aspect of the modern day human values including social, economic, political and religious. The long lineage of environmental degradation has been realized by the mankind at a huge cost where in the humanity is on the verge of facing its severest effects. The environmental debate and its related movements have passed through various phases like ecological, conservative, preservative, environment v. development and sustainable development. But the various diversified parameters, concepts, traditions and philosophies unified into a utilitarian and homogenous approach that is fundamentally concerned with the protection and management of the natural and human environment. Today the humanity has realized that nature is finite and that its misuse can ultimately threaten the human existence.

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The growth and expansion of 'environmental movement' has its omniscient impact on various fields including the form and format of its communication. The growth of personal, social and community awareness of environment has resulted in a surge of inter, intra and group communication about environment. In response, to this the mass media coverage of environment varied in many cycles directly reflecting the level of public interest and political action. Some periods witnessed peak media coverage followed by a lull due to the preoccupation of media with other important political or economic fallouts. The peak waves of environmental reporting was observed from 1961 to 1973, 1988 to 1991 and from 1999 to 2001. The peak wave started with from the early 1960's with the formation of World Wild Life Fund (1961) and the publication of the book 'The Silent Spring' written by Rachel Carson in 1962, widely credited for the beginning of the environmental movement in the world. The formation of

Friends of Earth (1969), establishment of Earth Day (1970), formation of green peace (1971), first United Nations conference on Human Environment held at stock home, formation of United Nations Environment Programme (1972) received wide media coverage. Disasters like Bhopal Gas tragedy (1984), Chernobyl disaster in Ukaraine, Rhine disaster in Europe (1996) started the second peak wave. Issues like ozone depletion and rainforest destruction along with the new movement of 'Green consumerism' dominated the second peak wave from 1988 to 1991, followed by the agenda of Sustainable development, Global governance agenda in the third peak wave from 1999 to 2001. There is actually a surge in environmental reporting throughout the world with the publication of the fourth report of the Intergovernmental Panel on Climate Change.¹

According to Max Boykoff, the climate news coverage has become a regular feature in English-language media in India. The volume of all climate change stories in the English Language Indian print media coverage suggests that it peaked in 2007 and then again around the time of summits in Copenhagen and Cancun.² But the ostrich like attitude of people as well as media towards the environmental issues resulted in prioritization of environment news without having a permanent approach. The traditional media with their agenda setting approach are more concerned and engaged with the ratings driven maneuvers like political ramblings and sitcom reruns. The environmental news reporting has tended to focus more on pollution incidents, ecological disasters and accidents etc rather than on highlighting social, economic causes and solutions. This attitude keeps news columns busy with specific and particular incidents. For example the issue of big dams in 1995, power projects in 1996, wild life in 1999, Global warming from the year 2000 was the main issue in Indian media columns.

Varied forms of environment communication like nature writing, environmental literature, environmental interpretation and environmental advocacy have found space and time in the media coverage. The mass media apart from creating awareness, evoking interest,

stimulating debate against environmental abuse has also abridged the 'inter-continental' gap, linking up all the people of the world to a common cause. International environmental news today receives as much sometimes more attention than the local environmental news. Environmental information has become more sophisticated and today, various ecological concepts like depletion of ozone layer, nuclear contamination, global warming, climate change etc. have become common vocabulary in journalistic as well as public parlance. Thus mass media can be given credit for bringing the environmental issues into public discourse and debate.

However, increased scientific debates and its communication have not translated into a public consensus. For example, Gallup Polls, trends indicate that there is a substantial decline in the percentage of Americans believing that global warming is already occurring i.e. from 61% to 50% from 2008 to 2010. The same trend was observed in the citizens of the other developed nations. This high level of skepticism and the decline in public acceptance of Climate Change call for increased examination of the factors influencing public opinion on Climate Change. The assumption that more information will lead to greater public acceptance of the reality greater support for environment policies is overly simplistic and trivialized. Thus, a framing analysis especially on a national, regional and local scale on the type of information disseminated on environmental issues and its impact on policy discourse and public understanding of environment was found necessary.

Framing analysis

The aspects of media coverage and the public understanding of environment are technically dealt in the Circuits of communication model developed by Carvalho and Burgess³. This model illustrates three phases or circuits through which communications originate and disseminate into the public sphere before entering the private sphere. The first phase involves news production and framing. Media professionals like editors and reporters produce news within a public and personal spheres of influence involving political, economic,

institutional, social and cultural landscape. News production also gets influenced by journalistic norms and values. In the second phase, the environmental texts in the form of news stories, opinion pieces etc are disseminated as encoded messages into public sphere through various mass media agencies. These climate news stories compete with other issues for public attention in the public sphere. The third phase examines the knowledge transfer and personal engagement of citizens with the environmental issues via the mass media. In this phase, the public discourses percolate and integrate through various styles and degrees into personal understanding and behavior of individuals. The information disseminated through mass media is interpreted and translated into decisions that may result in potential behavioral change. This is a complex and dynamic process involving many opinion leaders and boundary spanners in the society. Thus, media coverage of various issues is not a compendium of news items and clips produced by reporters rather they represent key frames evolved through a subset of complex relationships between scientists, policy makers and the public often mediated through various personal, public and media agendas.

Media framing is regarded as an important element of the news production phase of the model. In this the news makers i.e. media professionals engaged in the news collection, preparation and writing process construct the news. Media frames are defined as the central, organizing themes that connect different semantic elements of a news story into a coherent whole to suggest what is at issue. Essentially, frames are like maps that help people make sense of reality and are what people draw on to make decisions about what they should do during some social interaction. In social theory, a frame consists of a schema of interpretation, collection of anecdotes, and stereotypes that individuals rely on to understand and respond to events. In communication, framing defines how news media coverage can shape mass opinion by using these specific frameworks to help guide their reader to understanding.

Sociologist Erving Goffman described frames as a specific set of expectations that are used to make sense of a social situation at a given point in time⁴ The various elements of a news story like headlines, quotes, leads and its narrative structure all work together to organize or construct the meaning or the reality of the issue. Apart from them many other elements like emphasis, exclusion, elaboration and selection play a major role. Thus framing involves selection and salience i.e. to select some aspects of a perceived reality and make them more salient in a communicating text, in such a way as to promote a particular problem definition. A similar set of facts may be perceived quite differently when editors chose dramatically different media frames for stories.⁵ Thus news makers in the form of environmentalists, NGOs, activists etc try to promote their favored frames to media persons or news collectors. At the same time the media persons forge their own frames for various reasons of ideology, efficiency and story suitability.

Many types of frames like promotion v. prevention, now v. future, gain v. loss local frame be worried, public health solutions conflict social progress, conflict and strategy, scientifically uncertain, Pandora's box, public accountability, economic development, morality and ethics, public health etc are used in construction of environmental content by the news makers.⁶

Review of literature

In a study of news frames and European politics, Semetko and Valkenburg examined Dutch newspaper stories ($N=2,601$) and television news stories ($N=1,522$) from May 1 to June 20, 1997. The study examined the presence of five media frames that had been established from previous research, now known as generic frames. Generic frames were responsibility, human interest, consequences, morality and conflict. These frames commonly occur in media and transcend specific contextual factors such as issue coverage. The conflict frame accentuates conflict between individuals, groups or institutions as a means of capturing audience interest. The morality frame highlights the event, problem or issue in terms of the consequences, it will have

economically on an individual, group, institution, region or country. The responsibility frames presents an issue or problem in such a way as to attribute responsibility for its cause or solution to either the government or to an individual or group. The human interest frames brings a human face or an emotional angle to the presentation of an event, issue or problem. The economic consequences frame reports an event, problem, or issue in terms of the consequences it will have economically on an individual, group, institution, region or country. The responsibility frame was found to be most prevalent in Dutch news coverage, followed by frames of conflict, economic consequences, human interest, and morality.

Dirikx and elders⁷ analyzed Dutch and French newspaper news items ($N=257$) from four newspapers two from each country. These newspapers were published from 2001 until 2007 and focused on climate change. Results indicated that news items covering climate change focused on frames of consequences and responsibility. The consequences frame was the most prevalent frame. News items presenting the consequences frame frequently discussed the consequences of the pursuit of a certain course of action and of possible losses and gains. The second prevalent frame was the responsibility frame, in which news items discussed the need for urgent actions referenced possible solutions and suggested that governments are responsible for /or capable of alleviating climate change problems. The conflict frame and the human interest frame were the least prevalent frames according to the study.

Gordon, Deines and Havice⁸ analyzed newspaper coverage of climate change in a Mexico-city based newspaper called Reforma. Researchers examined news items ($N=144$) published from 2004 to 2006 to see how climate change was covered. Results indicated that newspaper coverage tended to rely on frames of consequences. Findings indicated an occasional presence of the conflict frame, but the conflict frame was not as prevalent as the consequences frame.

Jacobson, Langin, Carlton and Kaid⁹ investigated potential influences on public

support for the conservation of endangered large carnivores and found that newspaper coverage of the Florida panther had a neutral-to-positive frame valence. The content analysis examined newspaper news items ($N=513$) from 2003 to 2006 about the Florida panther and compared statewide and local news coverage. Local news coverage referred to reporting from local newspapers in Southwestern Florida, near a panther habitat. Results indicated that local newspapers devoted more coverage to panthers. While news coverage typically discussed associated risks, sources in news coverage were found to use a neutral-to-positive tone when speaking about panthers.

Framing analysis was conducted by Agwu Ekwe Agwu and Chiebonam Justina Amu of the Department of Agricultural Extension, University of Nigeria, Nsukka, Enugu State, Nigeria in 2013.¹⁰ The study examined the framing of climate change news in four national daily newspapers in Southern Nigeria namely Guardian, Vanguard, this day, and Daily Sun the countries leading national newspapers. The content analysis yielded approximately 332 climate change related news items. The results showed that greater proportion i.e. 125 out 332 news items used negative tone in reporting the headlines. Majority of news items i.e. 71 news items framed climate change in terms of blame, while action frames were used in 69 news items, among other frames. The study therefore recommends that media organizations should re-allocate some of their time and energy to explaining more of the specifics behind the mitigation and adaptive solutions to deal with global climate change, rather than devoting most of their time explaining the science behind global climate change.

AIMS AND OBJECTIVES

Considering the previous literature on framing analysis the following research questions were formulated :

RQ 1: What frames are used to portray environmental news items?

RQ 2: What frame / frames are dominant in the environmental news items?

RQ 3: Is there any shift in the frames or any predominant change over time?

RQ 4: Are the frames in the news items published in 2014 same as frames of news items published in 2004 and 2009 or different? This paper aims to understand the framing processes adopted by the regional media in disseminating information related to environmental issues. The paper also tries to understand how framing affects the coverage of the environmental issues. The idea of obtrusive and unobtrusive coverage was also taken into consideration while examining news stories relating to environment.

METHODOLOGY

The study intended to find out different forms of frames that newspaper stories adopted on the environmental issues. To analyze news stories for this purpose, content analysis method was chosen as it is the best suitable for gathering and analyzing the data from the stories. A quantitative framing analysis of news items published in the newspaper was used to answer the research questions. The sample was drawn from Eenadu Daily news paper from the year 2004, 2009 and 2014 (January to November). A total of 108 news items from 2004, 171 news items from 2009 and 432 news items from 2014 were used for quantitative framing. Every news item / news item was coded for a primary and secondary frame, the tone of the news item and other variables such as placing of the news item in the newspaper, length in column / centimeters (Col/Cms) and inclusion or absence of a visual element were also noted.

The search parameters stipulate that key words like climate change or global warming or greenhouse effect or alternate energy resources or Carbon Credits or Bio-Fuels or Natural Disasters or Natural Resources Management Green Technology must appear at least once within the news item. Different years were chosen so that the frames used to cover environmental issues could be analyzed for any shifts in framing. Each paragraph within the news item was examined for the appropriate frame and counted. The frame that was identified most often, the most prevalent frame, was coded as the primary frame. The primary frames identified by Boykoff were suitably adopted for the study. They are political frame, business and economic frame,

risk frame, public understanding frame, culture and religion frame, stewardship frame, science and technology frame, weather events frame, biodiversity frame, human impact frame etc.

Various Generic frames like responsibility, human interest, consequences, morality and conflict were used for the study. Other frames evolved from many other studies like frames involving the depth of coverage like Episodic, Thematic and Environmental Stewardship¹¹ was used for the study. The scientific frame i.e. uncertain science, controversial science etc are also used for the study.¹²⁻¹⁵ Inter-coder reliability was tested using another coder and it was found at 92% i.e. within the permissible result. Each story was also analyzed for the direction of tone : positive, negative and neutral.¹⁶⁻¹⁹ Tone of a story was based upon the number of sources in support of and against an issue.

HYPOTHESIS

The following hypothesis were formulated for the study :

H1: The depth of coverage of news items in 2014 is likely to be more thematic than the news items of 2004 due to the increased salience of environment news items.

H2: News items published in 2014 include more visual elements like picture, photograph, chart, Illustration, graph, etc than the news items that were published in 2004.

H3: There will be more analytical news items, editorials and expert opinions on environment when compared to news items of 2004.

H4: Environment news stories published in 2014 are more likely to have more regional / local frame than the news stories of 2004.

RESULTS AND DISCUSSION

The analysis of the articles in the year 2004 ($N=108$) indicates that the word pollution (19.44%) appeared in more stories followed by natural disasters (12.03%) and then by carbon credits (11.11%). The analysis of news item for the year 2009 ($N = 171$) reveal that the word climate change (17.54%) appeared in more stories followed by global warming (15.78%), then with natural disasters (14.62%) and alternate energy resources (14.03%) The analysis of news items of the year 2014 ($N=423$) indicate that the word climate change (28.13%) appears predominantly, followed by

natural disasters (16.31%) and alternate energy resources (14.89%). The analyses of the three years as seen in Table 1, Which indicates that

the quantum of news items which the reference of word climate change increased from 1.8% in 2004, 17.54% in 2009 and to 28.13% in 2014.

Table 1 : Classification of news items according to the key words

Key Words	2004	2009	2014
	Number (Percentage)	Number (Percentage)	Number (Percentage)
Climate change	2 (1.8%)	30 (17.54%)	119 (28.13%)
Global warming	11 (10.18%)	27(15.78%)	58 (13.71%)
Greenhouse effect	12 (11.11%)	10(5.84%)	9(2.12%)
Alternate energy resources	9 (8.33%)	24 (14.03%)	63 (14.89%)
Carbon credits	12(11.11%)	11 (6.43%)	10 (2.36%)
Bio-fuels / alternate fuels	11 (10.18%)	4 (2.33%)	38 (8.98%)
Natural disasters / calamities	13(12.03%)	25 (14.62%)	69(16.31%)
Natural resources management	4 (3.70%)	8 (4.67%)	13 (3.07%)
Green technology	2(1.85%)	12 (7.01%)	14(3.31%)
Conservation / protection	10(9.25%)	9 (5.26%)	12 (2.83%)
Pollution	21 (19.44%)	8(4.67%)	13 (3.07%)
Environmental law / regulation	1 (0.92%)	3(1.75%)	5 (1.18%)
Total	108 (100%)	171 (100%)	423 (100%)

The trend indicates that there was a shift away from the term greenhouse effect and towards the term climate change. This trend is in lieu with the trends observed in various other studies and as indicated is due to the politicized connotation and poor scientific backup for the words global warming and greenhouse effect than climate change. The term greenhouse effect was present throughout the three time periods but was sharply decreasing. No significant trend was observed for the category of global warming. The findings were also consistent with those of Carvalho, who noted that after 1988 the term greenhouse effect began to slip from the media landscape. It was replaced by global warming,

which was the most popular term in 1990. Today the term Global warming is being replaced by Climate Change.

The analysis of news items according to the type as represented in Table 2, indicates that news stories predominate in all the three years. The news items are generally incident, event based and is much depended on the information provided by various sources to the media organizations. A sharp shift is observed in the quantum of analytical stories which have increased from 12.03% in 2004, 12.2% in 2009 to 27.8% in 2014. An increase in number of feature stories, editorials and expert interviews is also observed. This confirms and proves the Hypothesis 3.

Table 2 : Classification of news items according to formats

Type of news / formats	2004	2009	2014
	Number (Percentage)	Number (Percentage)	Number (Percentage)
News stories	79 (73.14%)	105 (61.4%)	144 (34.0%)
Analytical stories	13 (12.03%)	21 (12.2%)	118 (27.8%)
Feature stories	11 (10.18%)	29 (16.9%)	79 (18.6%)
Editorials	2 (1.85%)	8 (4.6%)	48 (11.3%)
Interviews	2 (1.85%)	6 (3.5%)	28 (6.6%)
Photo features	1 (0.926%)	2 (1.1%)	6 (1.4%)
Total	108 (100%)	171 (100%)	423 (100%)

A total of 702 news items i.e. 108 news items in the year 2004 and 171 news items in 2009 and 423 news items of 2014 were coded for conducting framing analysis. The primary frames analysis as represented in **Table 3**. The primary frame analysis of the 2004 news items (N=108) indicate that the Risk frame (17.59%) followed by Science and Technology frame (15.74%) and then by the political frame (13.88%) were more frequently observed than other frames. The analysis of the 2009 news items (N = 171) reveals that the Stewardship frame (18.12%) has evolved as the most predominant frame followed by weather events frame (15.78%) and then by public understanding frame (15.20%). The analysis of the news items of 2014 consolidates the trend of 2009. It reveals that the public understanding frame (19.14%) is the predominating followed by stewardship frame (18.67%) and weather events frame (17.73%). A drastic reduction in political frame is observed from 2004 (13.88%), 2009 (5.84%) to 2014 (2.12%). There is also a reduction in risk frame from 2004 (17.59%), 2009 (12.86%) to 2014 (12.53%). But the quality of

risk reporting has improved tremendously and is more substantive and effective. The weather events frame is substantiated as more and more disasters, natural calamities are being reported from all over the country and world.

The framing trends of the three study periods indicate that the Public Understanding frame, Weather events frame and the Stewardship frame are more statistically significant frames when compared with others. This indicates a trend towards portraying environment more as a social, cultural and societal issue rather than a scientific one. Most frequently a neutral frame was associated with regard to framing of science, followed by uncertain science frame. The regional media follows the national and international news papers in reporting science based stories and hence the ambiguous and contentious science cases must have been filtered through gate keeping methods. The frames that were used to portray environmental issues are changing over time. Although the risk frame and political frame were dominant in 2004, it is clear that there is a shift in reporting climate change as a scientific, social and community issue.

Table 3 : Categorization of news items in to primary frames

Type of frame	2004 (Percentage)	2009 (Percentage)	2014 (Percentage)
Political frame	15 (13.88%)	10 (5.84%)	9 (2.12%)
Business and economic frame	14 (12.96%)	15 (8.77%)	31 (7.32%)
Risk frame	19 (17.59%)	22 (12.86%)	53 (12.53%)
Public understanding frame	12 (11.11%)	26 (15.20%)	81 (19.14%)
Culture and Religious frame	2 (1.85%)	04 (2.33%)	12 (2.83%)
Stewardship frame	4 (3.70%)	31 (18.12%)	79 (18.67%)
Science and Technology frame	17 (15.74%)	24 (14.03%)	66 (15.60%)
Weather events frame	14 (12.96%)	27 (15.78%)	75 (17.73%)
Biodiversity frame	11 (10.18%)	13 (7.01%)	17 (4.01%)
Total	108 (100%)	171 (100%)	423 (100%)

The depth of coverage of news items in 2014 is likely to be more thematic than the news items of 2004 due to the increased salience of environment news items. The analysis of news items depth of coverage indicate that the news items published after 2004 i.e. in 2009 and

2014 are more thematic than the previous articles as indicated in **Table 4**. Upon examining through a chi-square test a statistical significance did exist in for articles published by a factor of .051, partially supporting the hypothesis.

Table 4 : Depth of coverage (N =702)

Depth of coverage	Before 2004 (n =108)	After 2004 (n =594)
	Number (Percentage)	Number (Percentage)
Episodic	45 (41.66%)	143 (24.07%)
Thematic	63 (58.33%)	451 (75.92%)

News items published in 2014 include more visual elements like picture, photographs, charts, Illustrations, graphs, etc than the news items that were published in 2004. The news items published in 2014 were having more visual elements especially photographs and illustrations. Charts, graphs and other forms of data presentation were not observed. A new media cohort termed as data journalism is still evolving in which number-based stories are published with interactive diagrams, charts, and maps. The regional media is still not equipped for such type of journalism. The hypothesis that Environment news stories published in 2014 are more likely to have more regional / local frame than the news stories of 2004 was disproved as the news items published in 2014 were having more national and international frame rather than local frame.

CONCLUSION

Communicating climate change and its consequences is a challenging task for the media in India especially for the regional media. Mass media coverage of environmental issues has been recognized as a key element in the policy discourse and public understanding of environment. As more and more narratives on climate change are being explored the framing of climate debate assumes more importance. The study indicates that environmental reporting is stabilizing and the media professionals are generating news items using various analytical tools. It also indicates that the media is more concentrating on real time effects and developments of climate change rather than on the political-policy battles. On the other hand, the study also implicates that the concept of risk reporting is still evolving and the language, conceptualization of risk is still filled with national and international anecdotes. The study

indicates a clear lack of regional frame which is making the climate change a far distant reality with disputed predictions. The lack of local frame in the regional media is limiting the effect of the environmental news on the readers. The readers fail to connect with the issues and instances from places devoid of proximity.

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